Labwork 6

Exercise 1:

a. Find the minimum of the function $f(x) = -x^3 e^{-x}$ on the interval $1 \le x \le 4$ by using the function `fminbnd` with the options:

```
options=optimset('Display','Iter','TolFun',1e-8);
```

b. How many iterations does it take to find the minimum of the function f?

c. Plot the graph of the function on the above interval to visualize the results.

Exercise 2:

a. Find the minimum of the function $f(x) = xye^{-(x^2+y^2)}$ by using the function `fminsearch`. Display the iterations and set the value 10^{-8} as the value for the `'TolFun'` option.

b. How many iterations does it take to find the results

c. Plot the graph of the function on the region $[-2,2]\times[-2,2]$ to visualize the results. Is there another value of (x, y) at which the function obtains its minimum value?

Exercise 3:

a. Find the root of the function $f(x) = e^{-x} - e^{-x^3} + 1$ by using the function `fzero`. Set the initial guess at $x_0 = 2$. Using the `optimset` function to display the iterations and set the value 10^{-4} as the value for the `'TolFun'` option. How many iterations does it take to find the root?

b. Plot the graph of the function on the region [0,1] to visualize the result. How does the map change if we change the upper bound to 5, 10, 100 and if we change the lower bound to -1, -5, -10?

Exercise 4: ODE equation

a) Find the general solution of the following first order ODE

$$\frac{dy}{dt} + y\tan(t) = \sin(2t)$$

b) Find the general solution of the following second order ODE:

$$\frac{d^2y}{dt^2} - \frac{dy}{dt} = 1 + t\sin(t)$$

Observe the number of arbitrary constants in the general solutions of the above two ODEs.

c) Find the particular solution of the following ODE:

$$\begin{aligned} \frac{dy}{dt} &= e^{-y} \\ y(0) &= 0 \end{aligned}$$

d) Find the particular solution of the following ODE:

$$\begin{aligned} \frac{dy}{dt} + Ky &= A + B\cos\left(\frac{1}{12}\pi t\right) \\ y(0) &= 0 \end{aligned}$$

where A, B and K are constants.

Exercise 5: EULER method

a) Solve the following ODE using your script and plot the solution along with the exact solution on the same plot using code ***yprime.m***

$$y' = t^2 + y, \quad 0 \le t \le 2$$
$$y(0) = 1$$
$$h = 0.25$$
$$y_{exact} = 3e^t - t^2 - 2t - 2$$

Code **yprime.m**

```
% ODE
function yprime = yprime(t,y)
yprime = t^2 + y;
```

b) Compute solutions to the ODE in the above example for h = 0.5, 0.25, 0.1, and 0.01 with all other parameters remaining the same. Plot these four

solutions along with the exact analytical solution on the same graph. Adjust line styles using the figure menus.

CODE from `forwardEuler.m`

```
function [T,Y] = forwardEuler(yprime,tspan,h,y0)

% function [T,Y] = forwardEuler(yprime, tspan,h,y0)
% Computes the numerical solution of an ODE using the forward Euler method.
% yprime: handle of function defining the ODE.
% tspan = [tmin tmax] : interval over which solution has to be computed
% h : time step
% y0 : initial condition
% ------------------------
% Output:
% T: sequence of time steps
% Y: values of y at each time step

% Initializing our interval with the first time value:
T(1) = tspan (1);

% Initializing our solution with the initial condition or guess:
Y(1) = y0;

% Initializing n:
n=1;

while T(n) < tspan(2)
% Advancing time by the input time step:
    T(n+1) = T(n) + h ;
% Evaluating the solution at the next time step:
Y(n+1) = Y(n) + h* yprime(T(n),Y(n)) ;
% Incrementing n:
n=n+1;
end
```